Greenfield Institute

Course Registration System - Three-Tier Architecture Implementation
Submission Date:

👥 Group 14 - Team Members

Name	Registration Number	Role in Project
Kelvin Mutinda	SCM211-0325/2024	Admin Pages (Dashboard, Course Management, Registrations)
Ida Gakii	SCM211-0364/2024	Student Pages (Dashboard, My Courses, Course Browsing)
Natasha Hinga	SCM211-0311/2024	Backend Development (Database Connection, Session Management)
Victor Ngatia	SCM211-0117/2023	Backend Development (Course Registration Logic, XML Integration)
Vivian Leah Muthoni	SCM211-0340/2024	Login and Registration System (Authentication, Form Design)

📋 Project Overview

Greenfield Institute, a mid-sized educational organization, needed a modern web-based course registration system to replace their manual email and spreadsheet-based process. This system streamlines registration, improves data accuracy, and enhances user experience for both students and administrators.

🏗 Three-Tier Architecture Implementation

1. Presentation Tier (User Interface)

Technologies: HTML5, CSS3, JavaScript
Purpose: Provides visual interface for users to interact with the system.
Implemented by: Vivian Leah Muthoni (Login/Register), Ida Gakii (Student Pages), Kelvin Mutinda (Admin Pages)
Key Files:

- `login.php` - Student/Admin login with role toggle
- `register.php` - Student registration with validation
- `student_dashboard.php` - Student course browsing and enrollment
- `admin_dashboard.php` - Admin course management interface
- `my_courses.php` - View and drop enrolled courses
- `css/student_dashboard.css` - Responsive styling
- `js/my_courses.js` - Search, filter, modal interactions

2. Business Logic Tier (Application Layer)

Technologies: PHP 8.x
Purpose: Processes user requests, enforces business rules, and coordinates data flow.
Implemented by: Natasha Hinga & Victor Ngatia
Key Files & Functions:

- `config/database.php` - MySQL connection management
- `includes/functions.php` - Session handling, authentication, sanitization
- `register_course.php` - Prevents duplicate registrations, checks capacity
- `drop_course.php` - Removes course enrollment
- `add_course.php` / `edit_course.php` / `delete_course.php` - Admin course operations
- `import_xml.php` / `export_xml.php` - XML data exchange

Business Rules Enforced:

- ✅ No duplicate course registrations per student
- ✅ Course capacity limits (cannot exceed capacity)
- ✅ Password hashing for security (password_hash, password_verify)
- ✅ Role-based access control (Student vs Admin)
- ✅ Cannot delete courses with enrolled students

3. Data Tier (Database Layer)

Technologies: MySQL 8.0
Purpose: Stores and manages all persistent data.
Implemented by: Natasha Hinga
Database Schema:

Table	Columns	Purpose
`users`	id, name, email, password_hash, role, student_id, created_at	Stores student and admin accounts
`courses`	id, course_code, course_name, description, instructor, capacity, semester	Stores all course information
`registrations`	id, user_id, course_id, registration_date, status	Links students to courses (many-to-many relationship)

Key Relationships:

- Foreign key: `registrations.user_id → users.id` (ON DELETE CASCADE)
- Foreign key: `registrations.course_id → courses.id` (ON DELETE CASCADE)
- Unique constraint: `(user_id, course_id)` prevents duplicate registrations

🔄 How Data Flows Through the Tiers

```
Example: Student registering for a course

1 Presentation Tier: Student clicks "Register" on student_dashboard.php
↓
2 Business Tier: register_course.php checks for duplicates & capacity
↓
3 Data Tier: MySQL queries check existing registrations, then INSERT new record
↓
4 Response: Success/error message sent back to Presentation Tier
```

📄 XML Integration

The system demonstrates XML for structured data exchange:

- **Export:** `export_xml.php` - Converts MySQL course data to XML format for download
- **Import:** `import_xml.php` - Parses XML files using PHP's SimpleXML and inserts into database
- **Sample XML:** `xml/courses.xml` - Example course data structure

✅ Features Implemented

- ✅ Student registration with secure password hashing
- ✅ Student/Admin login with role-based redirection
- ✅ Browse available courses with real-time capacity display
- ✅ Register for courses (duplicate & capacity prevention)
- ✅ View enrolled courses with progress indicators
- ✅ Drop courses with confirmation modal
- ✅ Admin dashboard with statistics (students, courses, registrations)
- ✅ Add, edit, and delete courses (delete only if no students enrolled)
- ✅ View all student registrations in one table
- ✅ XML import/export functionality
- ✅ Responsive design (mobile compatible)
- ✅ Client-side form validation (JavaScript)
- ✅ Server-side validation & sanitization (PHP)
- ✅ SQL injection prevention (prepared statements)
- ✅ Search and filter courses

📊 Sample Data

Courses (8 courses):

Code	Course Name	Instructor	Semester
SMA3101	Probability and Statistics III	Prof. James Otieno	Semester 1, 2026
SMA2100	Introduction to Abstract Algebra	Dr. Mercy Wanjiku	Semester 1, 2026
ICS2402	Internet Application Programming	Prof. Kamau Maina	Semester 1, 2026
ICS2305	System Analysis and Design	Dr. Achieng Omondi	Semester 2, 2026
BUS1102	Entrepreneurship Skills	Prof. Njeri Mwangi	Semester 2, 2026
SMA1104	Linear Algebra I	Dr. Odhiambo Oduor	Semester 1, 2026
ICS2312	Operating Systems II	Prof. Wanjiru Kariuki	Semester 2, 2026

ICS2405	Statistical Programming	Dr. Atieno Ochieng	Semester 2, 2026

Test Credentials:

Role	Email	Password
Admin	admin@greenfield.edu	admin123
Student	john.mwangi@student.edu	student123
Student	mary.wanjiku@student.edu	student123

🎯 Conclusion

The Greenfield Institute Course Registration System successfully addresses the institution's registration challenges by providing a centralized, web-based solution. The three-tier architecture ensures:

- **Scalability** - Each tier can be upgraded independently
- **Maintainability** - Clear separation of concerns
- **Security** - Password hashing, prepared statements, session management
- **Usability** - Responsive design, intuitive interface

All assignment requirements have been met, including the three-tier architecture, XML integration, and all core features for both students and administrators.

Greenfield Institute of Science and Technology